

Omer Raza

MACHINE LEARNING ENGINEER

 Website  Scholar
 Github  LinkedIn

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Professional Summary

Machine Learning Engineer with 3+ years of experience deploying end-to-end ML systems in healthcare and logistics. Scaled CV, OCR, and risk models on AWS, via automation and CI/CD. Strong research foundation (EMNLP, IEEE) in GANs and LLMs; complemented by full-stack skills (React) to deliver production-ready solutions. Recognized with top-tier performance award.

Skills

Machine Learning: Computer Vision (OCR, GAN, Yolo & OpenCV), LLM (Llama, Qwen), XgBoost, GNN

Deep Learning Frameworks: Pytorch & Hugging Face (extensive), Scikit-learn (extensive), TensorFlow (familiar), JAX (familiar)

Data Processing: Spark, SQL, Pandas, Numpy

Cloud & MLOps: AWS, Kafka, GitLab CI/CD, Docker

Programming Languages: Python, JavaScript, C++

Web & Mobile: React & Electron

Industry Experience

Lalamove, Hong Kong - *Multinational delivery logistics company (10M+ users, 100K+ daily transactions)* **Feb 2022 - May 2023**

Machine Learning Engineer

Services & MLOps tools: Lambda API, EC2, Sagemaker, Real time/Batch processing & Kafka.

- Led accuracy boost from 65% to 85% in brand vehicle detection for market penetration tracking by standardizing label schema across teams, applying targeted augmentations (sticker masking) and model decoupling – saving \$70K+ annually
- Developed and deployed object detection, OCR (with clustering heuristics) via automated evaluation and CI/CD (GitLab + AWS Lambda with Docker images), cutting driver bonus approval time from 2 days to 5 seconds.
- Engineered a risk assessment model (F-score: 0.95) by implementing a full pipeline (data ingestion to ETL to feature engineering to real-time inference). This boosted daily fraudulent transaction detection by 20%.

Received the highest performance award category in the year (4 people among 150 in the office)

Gense Technologies, Hong Kong - *Award winning medical imaging (EIT) healthcare startup*

Nov 2020 - Dec 2021

Machine Learning & Software Engineer

- Deployed the Gense Mobile App (React) with live syncing to a proprietary EIT device, supporting user account and EIT test session management. This enabled rapid internal testing and real-time demos for partner investors.
- Developed iterative experimentation-driven ML models, achieving 88% EIT image quality and 99% breathing alignment accuracy, which enabled confidence-based diagnostics in the app.

Research Experience

RA Machine Learning at CIVS (Purdue University) on multimillion dollar grants for steel industry

Sep 2023 – June 2025

- Architected a silicon content prediction pipeline for blast furnace (90% accuracy; beta testing at partner plant) to improve production efficiency. Applied SHAP and expert-vetted features to provide interpretable, decision-relevant model insights.
- Developed a hearth erosion model (beta testing soon) by implementing first-principle models and optimization techniques (3-5 min), later accelerated with GNNs (20 sec), which can improve operator safety and campaign life.

RA Machine Learning (remote) at CUHK and HKUST

Aug 2023 - Jul 2024

First/co-author of multiple ML papers in EMNLP & IEEE on GANs & LLMs (see website for publications). Select papers:

1) FIRST: Teach a reliable large language model through efficient trustworthy distillation – Coauthor - EMNLP 2024

- Efficient LLM training via distillation with trustworthy maximization. (2.3% Accuracy ↑ and 10% miscalibration ↓)

2) SR TGAN: Smoke Removal with Temporal GAN in Robot assisted surgery – Equal Contribution First Author – IEEE 2024

- Incorporated sequential information from temporal video frames via attention, convolution & LSTM mechanisms. Better smoke removal (5-10%) than benchmark DeSmoke CycleGAN model on the JNBM & FADE metrics.

Independent Research Project - Facial Expression Transformation via Latent Embedding Interpolation

2021

- Automated large-scale (10K+) comic panels scraping and face extraction via custom scripts and CNN detection.
- Analyzed latent space interpolation in GANs, including StyleGAN, for facial transformation tasks (e.g stoic → expressive).
- Findings: Hyperplane-guided vector shifts led to stabler generative outputs than direct transformation models.

Education

Masters in Computer Science at Purdue University, GPA 4.0/4.0

Aug 2023 - May 2025

Bachelors in Computer Science at The University of Hong Kong (HKU), First Class Honors

Aug 2016 - Jun 2020

1) Deans Honor List, 2) HKU Foundation Scholar & 3) Young Tsun Dart Scholar (1 student / year / department)